Retrieval Augmented Generation (RAG) - Overview

Retrieval Augmented Generation is the process of optimizing the output of a large language model by referencing an authoritative knowledge base outside of the model's training data before generating a response. RAG extends the capabilities of LLMs to specific domains or an organization's internal knowledge base without retraining the model, making it a cost-effective way to keep answers relevant, accurate, and useful.

A RAG system consists of two pipelines: a data injection pipeline (parse documents, chunk them, embed the chunks, store vectors in a vector database) and a retrieval pipeline (embed the user query, run similarity search, pass the retrieved context plus a prompt to the LLM to generate the final answer).

Chunking Strategies

Chunking divides documents into smaller pieces that fit within the context size of embedding models and LLMs. The recursive character splitter tries a hierarchy of separators - paragraphs, then lines, then words - so chunks break at natural boundaries. Chunk overlap preserves continuity between neighboring chunks so information at the boundary is not lost.

RAG reduces hallucination: when the LLM is instructed to answer only from retrieved context, questions about content missing from the knowledge base can be answered with 'no relevant context found' instead of a fabricated response.